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Spectral-Spatial Context-Aware Target Prototype Learning for Hyperspectral Target Detection

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***Abstract*—Detecting targets of interest in hyperspectral images (HSIs) plays a significant role in Earth observation applications. In recent years, it has witnessed the development of deep learning methods in the field of target detection in HSIs. However, target detection still faces a formidable challenge, that is, spectral variation, especially the spectral variation of targets. The spectral variation of targets will lead to inconsistent semantic expressions for target pixels in deep features, making the output response of target pixels significantly different, thereby resulting in false detections and poor detection performance. To address this issue, we propose a new spectral-spatial context-aware target prototype learning method (denoted as S^{2}CTPL) for hyperspectral target detection. To capture discriminative spectral and spatial features from the image, a spectral-spatial residual extraction module is investigated and designed. In addition, we construct a multi-directional spatial context-aware module to model long-range dependencies of the spectral sequence, as well as global contextual dependencies of the HSI patches. Furthermore, to alleviate the inconsistent semantic expressions for target pixels, a new prior-guided target prototype learning network is proposed, in which a learnable prototype memory bank is used to provide more stable and robust representations of various target pixels. Finally, the detection result is achieved by the learned prototype memory bank and pixel patches. Extensive experiments are conducted on five benchmark detection datasets, and the results demonstrate that the S^{2}CTPL method achieves excellent detection performance in HSIs, surpassing the state-of- the-art methods.**

***Index Terms*—Hyperspectral image, target detection, spectral spatial joint, prototype learning.**

I. Introduction

HYPERSPECTRAL imaging has become a critical technology in modern remote sensing due to its capability to acquire dense spectral measurements across hundreds of contiguous wavelength bands [1], [2], [3], [4], [5]. Unlike multispectral imagery, hyperspectral images (HSIs) provide detailed spectral signatures that encode rich information of materials [6]. Such fine-grained spectral information enables the discrimination of targets with subtle differences, thereby offering significant advantages in numerous application fields, including environmental monitoring [7], classification [8], resource exploration [9], and military reconnaissance [10]. Among these tasks, hyperspectral target detection (HTD) has emerged as a fundamental yet challenging problem, aiming to accurately identify target materials of interest from complex and cluttered backgrounds [11], [12], [13].

Hyperspectral target detection (HTD) aims to identify and localize materials of interest within a scene by leveraging their spectral characteristics [2], [14]. Different from conventional salient object detection methods that typically depend on bounding boxes to localize distinct objects (e.g., vehicles or airplanes), HTD conducts detection at the pixel level, where each pixel is individually evaluated and assigned to either the target or background category [15], [16], [17], [18]. This formulation essentially casts HTD as a binary decision problem over all pixels in the image [19], [20]. However, directly applying traditional classification paradigms to HTD is not always appropriate. In HSIs, target pixels usually account for only a very small fraction of the data, resulting in severe class imbalance[21]. Under such conditions, overall classification accuracy can be dominated by background samples, which may lead to suboptimal detection performance [22]. Furthermore, most HTD approaches are implemented in a supervised manner and rely on prior knowledge of the target, typically provided in the form of one or a few given spectral signatures, to guide the target detection process [23], [24].

Over the past decades, extensive efforts have been devoted to HTD, and existing methods can be broadly categorized into three groups.

1) *Hypothesis model-based methods*: Early HTD methods are mainly based on statistical hypothesis testing, where target detection is performed by evaluating the likelihood of each pixel belonging to the target class. Representative methods include CEM [25], ACE [26], and OSP [27]. To improve robustness, various extensions have been developed, such as subspace modeling, regularization, kernel-based detectors [28], [29], [30], and fusion strategies [13], [31]. Although these methods are theoretically interpretable, their effectiveness relies heavily on background assumptions. In real-world HSIs, complex environments and nonlinear spectral mixing often violate these assumptions, leading to limited performance.

This work was supported by the Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Min-istry of Education of China under Grant JYB2025XDXM101, the National Natural Science Foundation of China under Grant 42501477, the New Cornerstone Science Foundation through the XPLORER PRIZE, and the Innovative Research Group Project of Hubei Province under Grant 2024AFA017. (*Corresponding authors: Rong Liu*)

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2) *Machine learning-based methods*: To overcome the limitations of hypothesis-driven detectors, various machine learning-based HTD methods have been developed [32], [33], [34]. Unlike statistical approaches, these methods adopt data-driven modeling to learn more discriminative representations or similarity measures from HSIs [35]. Representative techniques include sparse and collaborative representation models [36], metric learning methods [37], and low-rank decomposition approaches [38]. Typical examples are STD [39], SRBBH [40], and CSCR [41], which exploit the self-expression property of HSIs, where each pixel can be approximated by a combination of its neighboring pixels. MMML [42] and MIML [43], which learn discriminative distance metrics for target-background separation; and SLRMD [44] and DSOSP [38], which decompose hyperspectral data into target and background subspaces. Despite machine learning-based methods improving flexibility, these methods still suffer from several limitations. Most of them rely on shallow models or feature transformations, which restrict their ability to capture complex nonlinear relationships in hyperspectral data. In addition, their performance is sensitive to the quality of training samples and dictionary construction, which is particularly challenging under limited supervision.

3) *Deep learning-based methods*: The rapid advancement of deep learning has significantly promoted the development of hyperspectral target detection, driving the field from shallow feature modeling toward hierarchical representation learning [45]. Benefiting from the powerful feature extraction capability of deep neural networks, these methods can jointly exploit spectral signatures and spatial contextual information, thereby learning more discriminative representations for distinguishing targets and backgrounds. Compared with statistical and machine learning-based approaches, deep learning methods are generally more effective in capturing nonlinear spectral variations and high-level semantic characteristics embedded in HSI. From the perspective of training strategies, existing methods can be broadly categorized into supervised approaches and unsupervised frameworks [46], [47]. Supervised approaches construct end-to-end detection frameworks using labeled samples or synthesized target-background pairs. By explicitly modeling the relationships between target and background pixels, these methods learn discriminative feature mappings that facilitate target-background separation in the learned feature space. Representative methods include HTD-Net [48], TSCNTD [49], STTD [50], SASN [51], TSTTD [52], and FAME-Net [53]. Despite their promising performance, the effectiveness of supervised methods largely depends on the availability and quality of annotated samples, which are often difficult and costly to obtain in practical applications. In contrast, unsupervised methods mainly rely on representation learning without requiring explicit target labels. A common strategy is to employ deep autoencoder architectures to learn latent features that capture the intrinsic structure of HSIs. Target detection is then performed based on reconstruction residuals or feature discrepancies, under the assumption that background patterns can be well reconstructed while target pixels cannot. Such detectors include SRUN [54], ULMMDL [23], HTDIRN [55], SPDTT [56], and HTD-STF [57]. Beyond the two mainstream paradigms, several advanced deep learning techniques have been introduced to further enhance detection performance. Deep metric learning methods, such as DMLTD [58] and S^2D^3ML [59], aim to construct discriminative embedding spaces by learning similarity relationships between samples. Similarly, the deep contrastive learning method (e.g., DPN-CSSTD [60], MMSCL [61], and UEML [62]) alleviates the problem of poor separability between the target and the background in HTD through the representation learning mechanism of "pulling closer the positive samples and pushing away the negative samples". Moreover, Transformer-based architectures and attention mechanisms, such as CSTTD [63], GTPTN [64], LGA [65], and TCFSL [66], have demonstrated strong capabilities in modeling long-range dependencies and global contextual information, offering new opportunities for HTD.

Despite the remarkable progress, current HTD methods still face a critical challenge, that is, spectral variation, especially the spectral variation of targets. The spectral variation of targets will lead to inconsistent semantic expressions for target pixels in deep features, making the output response of target pixels significantly different, thereby resulting in false detections and poor detection performance. To mitigate this issue, some existing methods attempt to improve target representation by introducing multiple prior spectra. However, these approaches typically require a large number of high-quality prior spectra, and their performance is highly dependent on the quality and diversity of the priors. In practical applications, such prior spectra are often difficult to obtain, which limits the generalization ability of these detection methods.

In this context, prototype learning provides a promising alternative. The core idea of prototype learning is to represent each class using one or multiple prototypes, which serve as anchors in the feature space [67], [68]. By measuring the similarity between samples and these prototypes, the model can achieve robust classification or detection even with limited prior [69]. Therefore, prototype learning provides a natural way to handle spectral variability in HTD. By adopting prior spectra prototypes, it can capture diverse patterns within the same target class, effectively modeling intra-class variations. As a result, prototype-based representations can maintain both consistent semantic expressions and intra-class flexibility of target pixels, which are essential for reliable HTD.

Motivated by the above observations, this paper proposes a novel spectral-spatial context-aware target prototype learning based detector (S^2CTPL) for HSI target detection. In S^2CTPL, a novel prior-guided target prototype learning network is investigated, in which a learnable prototype memory bank that initialized with the prior spectra is used to provide more stable and robust representations of various target pixels. In this case, it effectively mitigates the impact of target spectral variability, and then enhances the semantic expression consistency of target pixels in deep features, thus leading to a competitive target detection performance in HTD. The main contributions of this work are summarized as follows:

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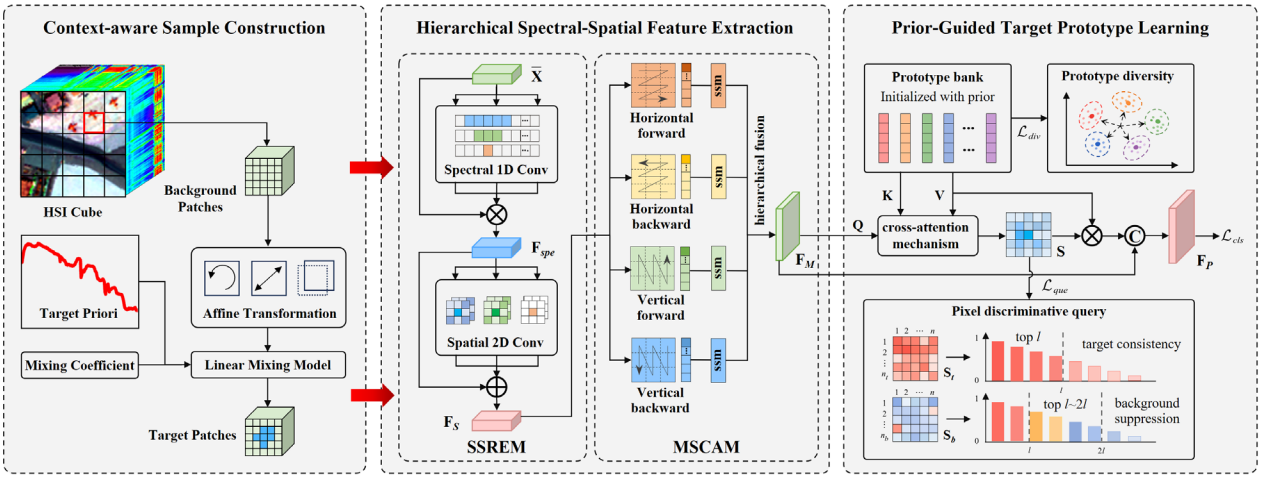


Fig.1. The flowchart of the S^{2}CTPL method for HTD.

1) A new target detection method called S^{2}CTPL is proposed. In the S^{2}CTPL, we investigate a new prior-guided target prototype learning network to alleviate the inconsistent semantic expressions for target pixels, in which a learnable prototype memory bank is used to provide more stable and robust representations of various target pixels.
2) To improve discriminative representation learning for HSI, a spectral-spatial context-aware feature extraction framework is presented, in which a spectral-spatial residual extraction module is to capture discriminative spectral and spatial features, while a multi-directional spatial context-aware module is to model long-range dependencies of the spectral sequence and global contextual dependencies of the HSI patches.

The remainder of the work is organized as follows. Section II presents the proposed S^{2}CTPL framework, including a detailed description of the core components and the overall methodology. Section III reports the experiments and provides comprehensive analyses on several benchmark hyperspectral datasets to evaluate the effectiveness of the proposed method. Finally, conclusions are drawn in Section IV.

II. The Proposed S^{2}CTPL Method for HTD

In this section, we present the spectral-spatial context-aware target prototype learning (S^{2}CTPL) framework for target detection. The overall architecture of S^{2}CTPL is illustrated in Fig.1, which mainly consists of three key components.

1) A context-aware sample construction strategy is first developed to alleviate the scarcity of target samples and simulate spectral variability by generating spatially and spectrally diverse target-background samples.
2) A hierarchical spectral-spatial feature extraction network is then constructed, including a spectral-spatial residual extraction module (SSREM) and a multi-directional spatial context-aware module (MSCAM), to capture both local discriminative features and long-range spatial dependencies.
3) Finally, a prior-guided target prototype learning network is proposed, where a learnable prototype memory bank is introduced to handle target spectral variability, and a discriminative query mechanism is designed to enhance target-background separability.

A. Context-aware Sample Construction

In deep learning methods for HTD, a major challenge stems from the extreme scarcity of target training samples. To address this issue, we propose a context-aware sample construction strategy to generate diverse training samples. Since targets in HSIs usually possess spatial continuity, the proposed sample construction strategy generates target samples in the form of pixel patches for subsequent training and detection.

Specifically, given an HSI $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, where H, W, and C denote the spatial dimensions and the number of spectral bands, respectively. The image is first partitioned into a set of patches $\mathbf{X}_b \in \mathbb{R}^{h \times w \times C}$, which are treated as background samples. To construct target samples, regions are randomly selected within the background samples as candidate regions for target implantation, where the pixel set of this region is $\boldsymbol{\Omega}$. To simulate the irregularity of real targets in HSIs and obtain more constructed samples, random affine transformations are applied to the candidate regions, including rotation, scaling, and translation operations:

$$f(\boldsymbol{\Omega}) = \mathbf{A}\boldsymbol{\Omega} + \mathbf{d} \tag{1}$$

where $\mathbf{A}$ represents the linear transformation matrix composed of rotation and scaling, and $\mathbf{d}$ is the translation vector. Through the above transformations, diverse target poses, scales, and spatial positions are introduced during the target sample construction process, thereby more realistically characterizing the spatial distribution properties of targets in complex scenes, obtaining the final target implantation region:

$$\bar{\boldsymbol{\Omega}} = f(\boldsymbol{\Omega}) \tag{2}$$

Furthermore, to simulate spectral variability under different background conditions and further enhance target samples, the linear spectral mixing model is adopted to generate target samples. This step linearly mixes the prior spectrum with the background spectrum within the implantation region and adds Gaussian noise to simulate spectral variability; the final target sample is constructed as follows:

$$\mathbf{X}_t(i,j) = \begin{cases} (1-\alpha)\cdot \mathbf{t} + \alpha \cdot \mathbf{X}_b(i,j) + \boldsymbol{\eta} & (i,j) \in \bar{\boldsymbol{\Omega}} \\ \mathbf{X}_b(i,j) & (i,j) \notin \bar{\boldsymbol{\Omega}} \end{cases} \tag{3}$$

where $\mathbf{t}$ is the prior target spectrum, $\alpha \in [0, 0.2]$ is the mixing

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coefficient that is set based on our previous work [49], and $\boldsymbol{\eta} \sim (0, \sigma^2)$ represents additive Gaussian noise.

By performing the above operations on each selected patch, a large number of spatially and spectrally diverse target samples are generated, which alleviates the class imbalance problem and improves the robustness of subsequent network training.

B. Spectral-Spatial Residual Extraction Module

To effectively extract discriminative features at different scales from samples, we propose a Spectral-Spatial Residual Extraction Module (SSREM).

First, a modified multi-scale spectral attention mechanism is proposed based on the ECA-net [70] to effectively model spectral dependencies. Existing spectral attention methods mostly rely on fully connected layers, ignoring cross-channel interactions important for continuous spectral signals. To address this issue, the fully connected layer is replaced with a multi-branch 1D convolution structure to capture spectral dependencies at different scales. Specifically, this mechanism uses 1D convolutions with different kernel sizes to process spectral vectors in parallel after integrating spatial information via a global average pooling operation.

$$\mathbf{z} = \frac{1}{h \times w} \sum_{i=1}^{h} \sum_{j=1}^{w} \bar{\mathbf{X}}(i,j) \tag{4}$$

$$\bar{\mathbf{z}}_r = conv1d_r(\mathbf{z}), r \in \{1, 3, 5\} \tag{5}$$

where $\bar{\mathbf{X}}$ represents the sample set composed of target samples and background samples, and $conv1d_r$ represents a 1D convolution with a kernel size of r. The output results from the multi-scale branches are concatenated along the channel dimension and fused with a convolution to achieve the aggregation of multi-scale spectral contexts, serving as weights indicating attention for different bands to recalibrate the original sample in the spectral dimension.

$$\mathbf{z}_{cat} = concat([\bar{\mathbf{z}}_1, \bar{\mathbf{z}}_3, \bar{\mathbf{z}}_5]) \tag{6}$$

$$\mathbf{F}_{spe} = \sigma(conv1d_1(\mathbf{z}_{cat})) \odot \bar{\mathbf{X}} \tag{7}$$

where $\odot$ represents a Hadamard product and $\sigma(\cdot)$ represents the sigmoid activation function.

This mechanism effectively models the correlation between local bands while maintaining the spectral dimension, highlighting more discriminative bands based on spectral context, avoiding the reduction of subsequent feature extraction efficiency due to information redundancy, and simultaneously maintaining the continuity of the original spectral signal.

To further enhance multi-scale spatial feature modeling capabilities, this work adopts a cascaded residual architecture to replace the traditional parallel inception structure. In the spatial feature extraction process, a 1×1 convolution is first applied to the input sample for spectral dimension compression to reduce feature dimensionality.

$$\mathbf{F}_{emb} = conv2d_1(\mathbf{F}_{spe}) \tag{8}$$

Subsequently, 1×1, 3×3, and 5×5 convolution operations are separately applied to the embedding feature to capture feature responses at different spatial scales. The outputs from different scale branches are concatenated in the channel dimension, then the spectral dimension is restored via projection convolution, and a residual connection is formed with the original input sample to obtain the spatial feature representation.

$$\mathbf{F}_{spa} = conv2d_1\left(\sum_{r=1,3,5} conv2d_r(\mathbf{F}_{emb}) \right) \tag{9}$$

where $conv2d_r$ represents a 2D convolution operation with a kernel size of $r \times r$, and r=1,3,5.

Then the final output of the SSREM can be calculated by

$$\mathbf{F}_S = \mathbf{F}_{spe} + \mathbf{F}_{spa} \tag{10}$$

Through the above operation, the SSREM can capture spatial-spectral features at different fine-grained levels of the training sample, thereby effectively improving the feature expression capability of HSI patches.

C. Multi-directional Spatial Context-Aware Module

Regarding the characteristic of high similarity among adjacent pixels in HSIs, convolution operations relying solely on local receptive fields are difficult to fully model the long-sequence spatial dependency relationships within HSI patches. Therefore, we designed a multi-directional spatial context-aware module (MSCAM) to model long-sequence spatial context dependencies in patches, thereby enhancing the spatial consistency and discriminative ability of features. The module implements improvements specifically for 2-D spatial image patches based on a state space model (SSM) [71] due to the powerful ability of SSM in modeling global contextual relationships across HSI patches.

Specifically, the feature $\mathbf{F}_S$ obtained from the SSREM is first reshaped and then embedded, including convolution layers, normalization layers, and activation functions.

$$\mathbf{E} = SiLU(BN(conv2d_1(\mathbf{F}_S))) \tag{11}$$

where $BN(\cdot)$ represents the batch normalization function, and $SiLU(\cdot)$ is the sigmoid linear unit activation function.

Standard SSM typically models 1D sequence data in a causal manner, which is unsuitable for non-causal high-dimensional hyperspectral data. Inspired by the Vision Mamba [72], this work designed a spatial scanning strategy and a spatial fusion mechanism to adapt to multi-dimensional spatial feature modeling requirements. To fully utilize the complementarity between spatial features, the embedded feature $\mathbf{E}$ is unfolded into four different 1D sequences through parallel paths. The four spatial scanning paths are defined as: horizontal forward scan, horizontal backward scan, vertical forward scan, and vertical backward scan.

$$\mathbf{X}_{seq}^{(1)} = Flatten(\mathbf{E}) \tag{12}$$

$$\mathbf{X}_{seq}^{(2)} = Flip(Flatten(\mathbf{E})) \tag{13}$$

$$\mathbf{X}_{seq}^{(3)} = Flatten(Transpose(\mathbf{E})) \tag{14}$$

$$\mathbf{X}_{seq}^{(4)} = Flip(Flatten(Transpose(\mathbf{E}))) \tag{15}$$

where $Flatten(\cdot)$, $Flip(\cdot)$, and $Transpose(\cdot)$ are the flattening, flipping, and transposition operations in the scanning path implementation process, respectively. $\mathbf{X}_{seq}^{(i)}$ represents the 1D sequence obtained from the i-th scanning path. Taking the horizontal forward scan as an example, the 2D embedding feature map is unfolded along the spatial dimension to generate a 1D sequence $\mathbf{X}_{seq}^{(1)} \in \mathbb{R}^{(hw) \times C}$. Subsequently, this sequence is

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input into the SSM module to model the long-sequence dependency relationships between pixels.

$$\mathbf{Y}^{(1)} = g(\mathbf{X}^{(1)}_{seq}) \tag{16}$$

where $g(\cdot)$ refers to the function of the SSM module, and $\mathbf{Y}^{(1)}$ is the sequence processed from the first path. The remaining three parallel paths adopt the same processing flow, but the scanning directions are different, and the corresponding model parameters are independent of each other.

After the above operation, the output sequences from the four paths are restored to the original spatial order, denoted as $\mathbf{Y}^{(i)}$, where $i \in \{1, 2, 3, 4\}$, corresponding to four scanning ways. To maintain spatial consistency while effectively fusing multi-directional information, a hierarchical feature fusion strategy is adopted: features from similar scanning directions (horizontal and vertical) are grouped and fused first to preserve axial spatial context information, and the results from horizontal and vertical processing are further combined for cross-directional interaction.

$$\mathbf{F}_{hor} = conv1d_1(concat[\mathbf{Y}^{(1)}, \mathbf{Y}^{(2)}]) \tag{17}$$

$$\mathbf{F}_{ver} = conv1d_1(concat[\mathbf{Y}^{(3)}, \mathbf{Y}^{(4)}]) \tag{18}$$

$$\mathbf{F}_{M} = conv1d_1(concat[\mathbf{F}_{hor}, \mathbf{F}_{ver}]) \tag{19}$$

Overall, the synergy between the SSREM and the MSCAM establishes a comprehensive hierarchical feature extraction framework. Crucially, SSREM acts as a foundational feature encoder for MSCAM. By filtering redundant spectral information and enhancing local spatial structures, SSREM transforms raw data into highly discriminative semantic tokens. This high-quality input empowers the subsequent Mamba block to more effectively model long-sequence dependencies without interference from noise, ensuring that the final feature representation possesses both precision and spatial consistency.

D. Prior-Guided Target Prototype Learning

To address the spectral variability of targets and maintain consistent semantic expressions for targets, this work proposes a prior-guided target prototype learning (PGTPL) network.

In this work, the targets are no longer modeled as single fixed spectra but are represented as a set of learnable target prototypes, denoted as $\mathbf{M} = \{\mathbf{m}_i\}_{i=1}^{n}$, where $\mathbf{m} \in \mathbb{R}^{1\times C}$ is a target-related distinguishable spectral representation, and n is the number of prototypes. By learning and maintaining a set of diverse target prototypes, it can effectively characterize highly similar and difficult-to-distinguish spectral profile features in the target spectrum, thereby achieving more robust and precise spectral analysis. And the target prototypes are initialized with target prior spectra, which can ensure that the model starts learning from a physically plausible starting point as well as accelerate convergence, and improve generalization performance. During the network training process, the prototype bank $\mathbf{M}$ is constantly updated and optimized.

First, a cross-attention mechanism is introduced to extract foreground target features. The target prototype set is viewed as a retrievable target memory bank. By calculating the semantic similarity between the sample spectral features $\mathbf{F}_M$ output by the MSCAM and the target prototype spectra, effective modeling of target-related information is achieved. Specifically, the aforementioned spectral features and the prototype memory bank are projected into Query (Q), Key (K), and Value (V) via learnable linear transformation matrices $\mathbf{W}^{Q}$, $\mathbf{W}^{K}$, and $\mathbf{W}^{V}$.

$$\mathbf{Q} = \mathbf{F}_M \mathbf{W}^{Q},\ \mathbf{K} = \mathbf{M}\mathbf{W}^{K},\ \mathbf{V} = \mathbf{M}\mathbf{W}^{V} \tag{20}$$

Subsequently, through the cross-attention mechanism, the attention weight matrix is calculated using the scaled dot-product attention form to quantify the similarity between sample features and the target prototype memory bank.

$$\mathbf{S} = \delta\left(\mathbf{Q}\mathbf{K}^{T} \Big/ \sqrt{d_k}\right) \tag{21}$$

where $\mathbf{S} \in \mathbb{R}^{(hw)\times n}$ that contains $\mathbf{S}_t$ and $\mathbf{S}_b$, $\delta(\cdot)$ represents the softmax activation function, and $\sqrt{d_k}$ is the scaling factor. Based on this attention weight, a weighted sum of the value vectors corresponding to the prototype memory bank is performed to obtain the target prototype-enhanced feature representation. This explicitly integrates the target semantic information learned in the prototype bank into the network representation, thereby reinforcing target discriminative features and improving the network's ability to distinguish the target from the background.

$$\mathbf{F}_{aug} = \mathbf{S}\mathbf{V} \tag{22}$$

Besides, it records the various possible variations that may occur in the prior spectra, thus addressing the spectral variability of targets and maintaining consistent semantic expressions for target pixels in deep features.

Finally, this enhanced feature is concatenated with the module input $\mathbf{F}_M$ in the channel dimension to form a syncretic feature to fully utilize the spectral-spatial feature and the enhanced features. The syncretic feature is further sent into the detection head for target discrimination. The detection head consists of two convolution layers, Top-k pooling, and two fully connected layers, finally outputting the target prediction probability $\hat{\mathbf{y}}$, which is used for the subsequent calculation of the loss function.

$$\mathbf{F}_P = concat([\mathbf{F}_M,\ \mathbf{F}_{aug}]) \in \mathbb{R}^{(hw)\times 2C} \tag{23}$$

$$\hat{\mathbf{y}} = \sigma\left(FC\left(pooling_k\left(conv1d_1(\mathbf{F}_P)\right)\right)\right) \tag{24}$$

where k represents the number of maximum response elements retained in Top-k pooling.

E. Loss Function

1) Binary Cross-Entropy Loss: The sample prediction score $\hat{\mathbf{y}}$ and the real labels corresponding to the artificially constructed target-background samples $\mathbf{y} \in \{0, 1\}$ are jointly used to calculate the target-background binary classification loss. Specifically, the Binary Cross-Entropy (BCE) loss function is adopted to supervise prediction results.

$$\mathcal{L}_{cls} = BCE(\mathbf{y}, \hat{\mathbf{y}}) \tag{25}$$

This loss function enables the network to possess preliminary discrimination capability by minimizing the difference between the predicted probability and the true label.

2) Pixel Discriminative Query Loss: To further enhance the ability to discriminate between targets and backgrounds, this work proposes a query loss function based on intra-sample pixel attribute differences. This loss guides the two to form a higher consistency representation by constraining the similarity

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relationship between target pixels and target prototypes, while effectively suppressing erroneous responses in backgrounds.

After passing through the PGTPL, the target sample and background sample obtain corresponding prototype similarity query score matrices $\mathbf{S}_t$ and $\mathbf{S}_b$. The element $s_{i,j}$ in this query score matrix represents the similarity between the i-th pixel in the sample and the j-th prototype in the prototype bank. A top-k supervision strategy is used on this similarity matrix to ensure that the model generates larger response differences for hard targets and backgrounds, thereby improving target-background distinctness. Specifically, for target samples, it is expected that there are at least l high-score prototypes highly similar to the input sample, meaning the top l values of the query scores corresponding to the target region pixels in the sample should approach 1. For background samples, ideally, all pixels should not match any target prototype, meaning the query scores of pixels with any target prototype should approach 0. To enhance the model's ability to distinguish difficult background pixels from target pixels, the query loss focuses on constraining and suppressing prototype responses with higher query scores. However, in practice, some hard background pixels may contain spectral components resembling the target due to mixed pixels or interference. Therefore, it is expected that the (l+1)-th to 2l-th high values $\hat{\mathbf{S}}_b$ in the query scores corresponding to background pixels approach zero. While suppressing significant non-target responses, this introduces a certain tolerance space for the target prototype learning process, allowing a small number of background pixels to generate limited responses to target prototypes, helping to improve the model's generalization ability under complex background conditions [73]. It can be expressed by the following formula.

$$\hat{\mathbf{S}}_t = select_{[1:l]}(\mathbf{S}_t(i)),\ i \in \bar{\mathbf{\Omega}} \tag{26}$$

$$\hat{\mathbf{S}}_b = select_{[l+1:2l]}(\mathbf{S}_b) \tag{27}$$

$$\mathcal{L}_{que} = BCE\left(\hat{\mathbf{S}}_b, \mathbf{0}\right) + BCE\left(\hat{\mathbf{S}}_t, \mathbf{1}\right) \tag{28}$$

where $\mathbf{S}_t(i)$ is the prototype query score corresponding to the target pixel in the target sample.

3) Prototype Diversity Loss: Since the prototypes tend to learn the same target features, leading the model into suboptimal solutions, an auxiliary prototype diversity loss is introduced. This guides different prototypes to focus on different spectral features by increasing the difference between different prototype features [74].

$$\mathcal{L}_{div} = \frac{1}{n(n-1)} \sum_{i=1}^{n} \sum_{j=1, j\neq i}^{n} \frac{\langle \mathbf{m}_i, \mathbf{m}_j \rangle}{\|\mathbf{m}_i\| \|\mathbf{m}_j\|} \tag{29}$$

where $\mathbf{m}_i$ is the i-th prototype in the target prototype bank, and n is the number of prototypes.

In summary, the total loss of the model is

$$\mathcal{L} = \mathcal{L}_{cls} + \lambda_1 \mathcal{L}_{que} + \lambda_2 \mathcal{L}_{div} \tag{30}$$

where λ_1, and λ_2 are the loss weights of three losses, and they are adjustable in the training process.

F. Detection Module

The detection module is divided into two stages, i.e., local patch-based detection and refined pixel-based detection.

First of all, during the local patch-based detection, patch scanning is performed on the hyperspectral image to preliminarily locate the target and suppress irrelevant backgrounds. The image is segmented into several image patches $\{\mathbf{P}_i\}_{i=1}^{N_P} \in \mathbb{R}^{h\times w\times C}$, where N_P is the number of image patches to be scanned, which can be calculated by the image size and the patch size. Each image patch obtains a corresponding target confidence score after passing through the aforementioned SSREM, MSCAM, and PGTPL. The target confidence scores of all image patches are sorted in descending order as follows.

$$A_{rank} = \left\{ (a_1, \mathbf{P}_1), (a_2, \mathbf{P}_2), \cdots, \left(a_{N_P}, \mathbf{P}_{N_P}\right) \right\} \tag{31}$$

where $(a_i, \mathbf{P}_i)$ is the i-th highest target confidence score and its corresponding image patch, and A_{rank} represents the score-image patch set sorted in descending order of target confidence scores. Image patches ranked at the top have a higher probability of target presence, while image patches ranked at the bottom are composed of background pixels. Based on this, the image patches with a confidence score over the predefined threshold of 0.1 are regarded as potential target regions, and the next step of pixel-wise fine verification is performed on them; all the remaining image patches are assigned as background.

In addition, refined pixel-based detection is conducted, and further pixel verification is implemented to refine detection results and reduce the false alarm rate. All pixels are sequentially selected from high-confidence image patches and implanted into random positions of the lowest-confidence image patch to constitute a new augmented image patch. Subsequently, the implanted patch is then re-fed into the network, and the obtained confidence score is used as the final detection result for the corresponding implanted pixel. By introducing pixels to be verified into a pure background context, this augmented image patch can effectively utilize high-confidence background information to distinguish target pixels from adjacent backgrounds. At the same time, image patches that might have been misjudged as potential targets in the first stage can be further discriminated through pixel-wise detection in this stage. Overall, the proposed two-stage detection strategy combines global coarse scanning with local fine verification, helping to achieve accurate separation and reliable detection of target pixels under complex backgrounds, and improves detection efficiency by performing pixel-wise verification in regions that are prone to being falsely detected.

In summary, the S^{2}CTPL detector conducts a spectral-spatial residual extraction module to capture discriminative features by integrating spectral and spatial information. To further enhance representation capability, a state-space modeling block is introduced to model long-range dependencies in spectral sequences as well as global contextual relationships across hyperspectral regions. In addition, a prototype learning network with a learnable memory bank is developed to provide stable and consistent representations for target pixels, effectively mitigating the impact of spectral variability. By jointly modeling contextual information and prototype-based representations, the proposed method achieves improved discrimination between targets and background samples, and thus contributes to good detection performance.

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TABLE I
COMPREHENSIVE INFORMATION OF FIVE HSI DATASETS. "-" MEANS THE INFORMATION IS UNKNOWN.

Dataset	Number of Bands	Image Size	Pseudo Color RGB	Spectral Resolution	Spatial Resolution	Target	Number of target pixels
Urban	204	100×100	64, 42, 30	-	17.2m/pixel	-	67
Sandiego	189	100×100	55, 35, 30	10nm	3.5m/pixel	airplane	134
Airport	205	100×100	99, 58,19	-	7.1m/pixel	airplane	87
Grand Isle	224	300×480	31, 22, 14	-	4.4m/pixel	man-made object	90
RIT Campus	360	200×200	55, 36, 18	5nm	1m/pixel	red panel	80

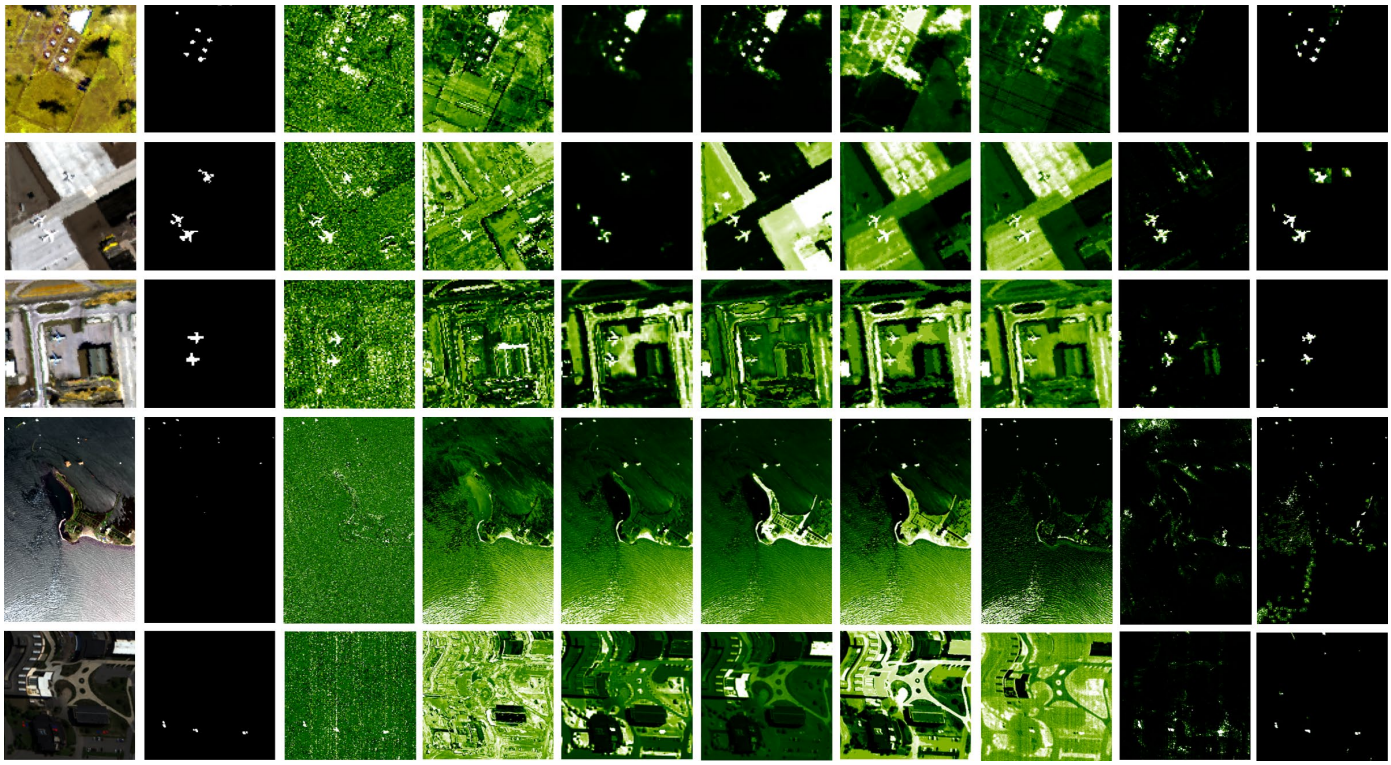


Fig.2. The depth detection map comparison for five HSIs. From top to bottom: Urban, Sandiego, Airport, Grand Isle, and RIT Campus.

III. EXPERIMENTS AND ANALYSIS

A. Hyperspectral Dataset Description

To evaluate the effectiveness of the S²CTPL, experiments are conducted on five widely used HSIs, including Urban [75], Sandiego [76], Airport [75], Grand Isle [77], and RIT Campus [78]. The pseudo-color images and ground truth are shown in Fig. 2. These HSIs cover a variety of complex scenes, including urban areas, airports, and coastal environments, with different spatial resolutions and spectral characteristics. The target types include airplanes, man-made objects, and specific panels, with the number of target pixels ranging from dozens to over one hundred, reflecting typical small-target detection scenarios in HSIs. Detailed information of all datasets, including the number of spectral bands, image size, spatial resolution, and target characteristics, is summarized in Table I.

B. Experimental Setup

This article conducts a comprehensive evaluation of the detection performance of the proposed S²CTPL framework by employing two widely used evaluation metrics, namely the ROC curve and the AUC value.

(1) ROC curve: In this work, the 3D ROC curve [79] is adopted, which is constructed as a function of the probability of detection (PD), false alarm rate (FAR), and a threshold parameter τ. By introducing τ as an additional independent variable, the 3D ROC curve provides a more comprehensive characterization of detector performance. From the 3D ROC surface, three types of 2D ROC curves can be derived, including the PD-FAR curve, the PD-τ curve, and the FAR-τ curve. Among these, the PD-FAR curve is primarily used to evaluate the overall detection performance, the PD-τ curve reflects the detectability under varying thresholds, and the FA-τ curve characterizes the background suppression capability.

(2) AUC value: To quantify the target detection performance, multiple AUC-based metrics are employed. Specifically, the area under the PD-FAR curve is first calculated and denoted as $AUC_{PD\text{-}FAR}$. Based on the 3D ROC formulation [79], two additional AUC metrics, namely $AUC_{PD\text{-}\tau}$ and $AUC_{FAR\text{-}\tau}$, are introduced. Building upon these fundamental AUC indicators,

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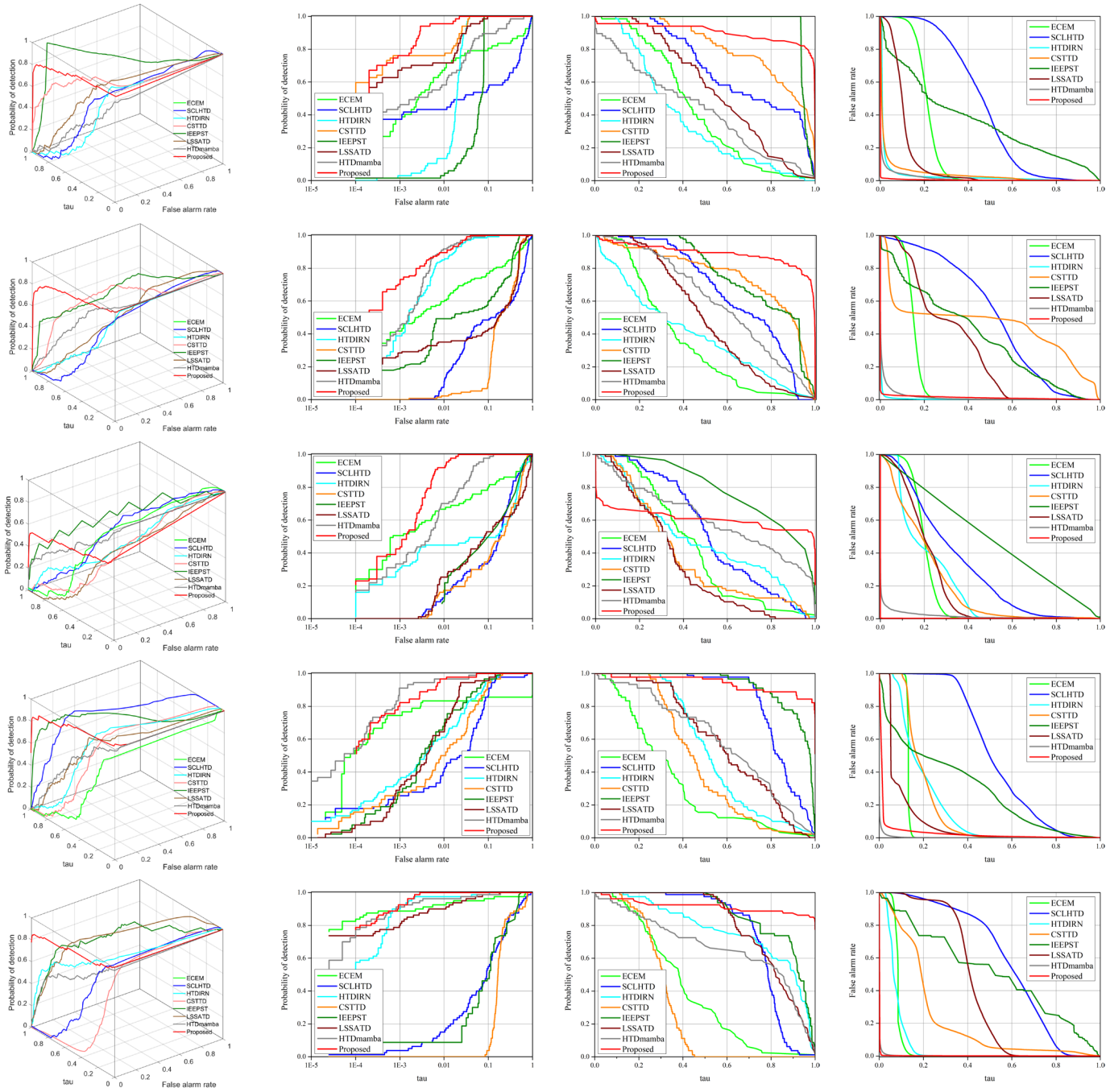


Fig.3. The comparisons of 3D ROC and 2D ROC performance of five datasets. From top to bottom: Urban, Sandiego, Airport, Grand Isle, and RIT Campus. From left to right: 3D ROC curve, 2D ROC curve of PD-FAR, 2D ROC curve of PD-τ, 2D ROC curve of FAR-τ.

several composite evaluation metrics are further derived, including the target detection capability AUC_{TD}, the background suppressibility AUC_{BS}, the joint evaluation of target detectability and background suppression AUC_{TD-BS}, the overall detection accuracy AUC_{OD}, the overall detection probability AUC_{ODP}, and the signal-to-noise probability ratio AUC_{SNPR}. These metrics collectively provide a comprehensive and multi-dimensional assessment of detection performance.

To validate the effectiveness of the proposed S^{2}CTPL method, extensive comparisons are conducted against seven state-of-the-art target detection methods, including ECEM [31], SCLHTD [80], HTDIRN [55], CSTTD [63], LSSATD [81], IEEPST [82], and HTDmamba [47]. It is worth noting that, to ensure a fair comparison, the same prior target spectrum is adopted for all compared methods. Meanwhile, the parameters of these methods are carefully tuned according to the recommendations in the corresponding literature to achieve their robust and reliable performance.

For the proposed S^{2}CTPL detector, two key parameters are the size of image patch $h \times w$ and the number of prototypes n, and they are empirically set to 7×7 and 128 in this work after extensive experiments. During the training stage, the number of image patches of target and background samples is 512, respectively. The deep network in the S^{2}CTPL is trained with

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TABLE II
AUC COMPARISON OF EIGHT DETECTORS FOR THE URBAN DATASET

Methods	AUC_{PD-PF}	$AUC_{PD-\tau}$	$AUC_{PF-\tau}$	AUC_{TD}	AUC_{BS}	AUC_{TD-BS}	AUC_{ODP}	AUC_{OD}	AUC_{SNPR}
ECEM	0.8757	0.4417	0.2127	1.3174	0.6630	0.2290	1.2290	1.1047	2.0766
SCLHTD	0.7522	0.7014	0.4579	1.4536	0.2943	0.2435	1.2435	0.9957	1.5318
HTDIRN	0.9811	0.3785	0.0282	1.3596	0.9529	0.3503	1.3503	1.3314	13.4220
CSTTD	0.9943	0.8184	0.0365	1.8127	0.9578	0.7819	1.7819	1.7762	22.4219
IEEPST	0.9399	**0.9523**	0.3582	1.8922	0.5817	0.5941	1.5941	1.5340	2.6586
LSSATD	0.9906	0.5705	0.1087	1.5611	0.8819	0.4618	1.4618	1.4524	5.2484
HTDmamba	0.9583	0.4278	0.0132	1.3861	0.9451	0.4146	1.4146	1.3729	32.4091
S^2CTPL	**0.9985**	0.9040	**0.0044**	**1.9025**	**0.9941**	**0.8996**	**1.8996**	**1.8981**	**205.4545**

TABLE III
AUC COMPARISON OF EIGHT DETECTORS FOR THE SANDIEGO DATASET

Methods	AUC_{PD-PF}	$AUC_{PD-\tau}$	$AUC_{PF-\tau}$	AUC_{TD}	AUC_{BS}	AUC_{TD-BS}	AUC_{ODP}	AUC_{OD}	AUC_{SNPR}
ECEM	0.8843	0.3588	0.1516	1.2431	0.7327	0.2072	1.2072	1.0915	2.3668
SCLHTD	0.7082	0.6759	0.5137	1.3841	0.1945	0.1622	1.1622	0.8704	1.3157
HTDIRN	0.9933	0.3891	**0.0039**	1.3824	**0.9894**	0.3852	1.3852	1.3785	**99.7692**
CSTTD	0.7335	0.7578	0.4468	1.4913	0.2867	0.3110	1.3110	1.0445	1.6961
IEEPST	0.8697	0.7779	0.3758	1.6476	0.4939	0.4021	1.4021	1.2718	2.0700
LSSATD	0.7694	0.5074	0.3078	1.2768	0.4616	0.1996	1.1996	0.9690	1.6485
HTDmamba	0.9958	0.6077	0.0137	1.6035	0.9821	0.5940	1.5940	1.5898	44.3577
S^2CTPL	**0.9968**	**0.8906**	0.0127	**1.8874**	0.9841	**0.8779**	**1.8779**	**1.8747**	70.1260

TABLE IV
AUC COMPARISON OF EIGHT DETECTORS FOR THE AIRPORT DATASET

Methods	AUC_{PD-PF}	$AUC_{PD-\tau}$	$AUC_{PF-\tau}$	AUC_{TD}	AUC_{BS}	AUC_{TD-BS}	AUC_{ODP}	AUC_{OD}	AUC_{SNPR}
ECEM	0.8935	0.4338	0.2000	1.3273	0.6935	0.2338	1.2338	1.1273	2.1690
SCLHTD	0.7805	0.5412	0.3171	1.3217	0.4634	0.2241	1.2241	1.0046	1.7067
HTDIRN	0.7664	0.4769	0.2053	1.2433	0.5611	0.2716	1.2716	1.0380	2.3229
CSTTD	0.7362	0.3809	0.2012	1.1171	0.5350	0.1797	1.1797	0.9159	1.8931
IEEPST	0.7910	**0.7903**	0.4932	1.5813	0.2978	0.2971	1.2971	1.0881	1.6024
LSSATD	0.7152	0.3319	0.2057	1.0471	0.5095	0.1262	1.1262	0.8414	1.6135
HTDmamba	0.9854	0.6077	0.0161	1.5931	0.9693	0.5916	1.5916	1.5770	37.7453
S^2CTPL	**0.9971**	0.6077	**0.0024**	**1.6048**	**0.9947**	**0.6053**	**1.6053**	**1.6024**	**253.2083**

100 epochs, a batch size of 64, and a learning rate of 0.0001. The loss weights λ_1 and λ_2 are set as 0.1. These hyperparameters are empirically determined to achieve stable convergence and robust optimization performance.

C. Comparison of Detection Performance

In this section, a comprehensive evaluation of the S^2CTPL is conducted by comparing it with several advanced detectors. The analysis is carried out from three perspectives, including qualitative visualization of detection maps, 3D ROC curve analysis, and quantitative comparison using AUC metrics.

Firstly, from a qualitative perspective, the detection maps of all compared methods across five hyperspectral datasets are illustrated in Fig. 2. It can be observed that traditional methods such as IEEPST and LSSATD generally suffer from high false alarm rates, especially in complex scenes such as Urban, Sandiego, Airport, and RIT Campus datasets, where large background regions are not effectively suppressed. Similarly, ECEM and SCLHTD exhibit insufficient background suppression, as a considerable number of background pixels remain highlighted. In contrast, deep learning-based methods such as HTDmamba demonstrate improved background suppression capability. However, their performance is still inferior to the proposed S^2CTPL method. From the perspective of target detection accuracy, SCLHTD shows relatively poor performance, as many target pixels are not correctly detected, especially in the Airport dataset (airplane targets) and the RIT Campus dataset (red panel targets). CSTTD also suffers from inaccurate localization in these datasets, leading to missed detections. In addition, ECEM and IEEPST exhibit inferior target detection performance in the Urban dataset, mainly due to their reliance on traditional filtering or distance measures, which fail to fully exploit deep spectral-spatial features, resulting in weak semantic consistency of target representation. In contrast, the S^2CTPL achieves significantly improved performance, where most background pixels are effectively suppressed while target regions are clearly highlighted. Moreover, the proposed method is capable of accurately delineating target boundaries, as evidenced in datasets such as Sandiego and Airport. These results demonstrate that the proposed method maintains strong semantic consistency for both background and target regions, leading to accurate and reliable detection performance across all datasets.

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TABLE V
AUC COMPARISON OF EIGHT DETECTORS FOR THE GRAND ISLE DATASET

Methods	AUC_{PD-PF}	$AUC_{PD-\tau}$	$AUC_{PF-\tau}$	AUC_{TD}	AUC_{BS}	AUC_{TD-BS}	AUC_{ODP}	AUC_{OD}	AUC_{SNPR}
ECEM	0.8540	0.3344	0.1314	1.1884	0.7226	0.2030	1.2030	1.0570	2.5449
SCLHTD	0.9475	0.8367	0.5297	1.7842	0.4178	0.3070	1.3070	1.2545	1.5796
HTDIRN	0.9788	0.5495	0.1909	1.5283	0.7879	0.3586	1.3586	1.3374	2.8785
CSTTD	0.9701	0.4729	0.1931	1.4430	0.7770	0.2798	1.2798	1.2499	2.4490
IEEPST	0.9828	0.9259	0.2938	1.9087	0.6890	0.6321	1.6321	1.6149	3.1515
LSSATD	0.9875	0.6063	0.0955	1.5938	0.8920	0.5108	1.5108	1.4983	6.3487
HTDmamba	**0.9982**	0.6035	**0.0029**	1.6017	**0.9953**	0.6006	1.6006	1.5988	**208.1034**
S^2CTPL	0.9978	**0.9412**	0.0282	**1.9390**	0.9696	**0.9130**	**1.9130**	**1.9108**	33.3759

TABLE VI
AUC COMPARISON OF EIGHT DETECTORS FOR THE RIT CAMPUS DATASET

Methods	AUC_{PD-PF}	$AUC_{PD-\tau}$	$AUC_{PF-\tau}$	AUC_{TD}	AUC_{BS}	AUC_{TD-BS}	AUC_{ODP}	AUC_{OD}	AUC_{SNPR}
ECEM	0.9800	0.3970	0.0845	1.3770	0.8955	0.3125	1.3125	1.2925	4.6982
SCLHTD	0.8375	0.7680	0.5916	1.6055	0.2459	0.1764	1.1764	1.0139	1.2982
HTDIRN	0.9979	0.7723	0.0762	1.7702	0.9217	0.6961	1.6961	1.6940	10.1352
CSTTD	0.7736	0.3014	0.2208	1.0750	0.5528	0.0806	1.0806	0.8542	1.3650
IEEPST	0.8474	0.8714	0.5110	1.7188	0.3364	0.3604	1.3604	1.2078	1.7053
LSSATD	0.9947	0.8040	0.4129	1.7987	0.5818	0.3911	1.3911	1.3858	1.9472
HTDmamba	0.9650	0.6701	**0.0020**	1.6351	0.9630	0.6681	1.6681	1.6331	335.0500
S^2CTPL	**0.9997**	**0.9163**	0.0027	**1.9160**	**0.9970**	**0.9136**	**1.9136**	**1.9133**	**339.3704**

Secondly, the detection performance is further evaluated through 3D ROC curves and the corresponding 2D projections, as shown in Fig. 3. In each subplot, the red curve represents the proposed S^2CTPL method. From the PD-FAR curves, it can be observed that the proposed method consistently maintains a leading position across all five datasets, especially in Urban, Sandiego, and Airport datasets. This indicates that the proposed method achieves higher detection probabilities under lower false alarm rates, demonstrating superior overall detection performance. From the PD-τ curves, it can be observed that although some competing methods may outperform the proposed method under very low threshold values, the S^2CTPL method exhibits clear advantages as the threshold increases, with its ROC curves generally located in the upper-right region. This demonstrates that the proposed method maintains strong target detectability across a wide range of thresholds. Furthermore, from the FAR-τ curves, it is evident that the proposed method consistently achieves the lowest false alarm rates among all compared methods. Specifically, most background pixels are effectively suppressed at relatively low threshold values, and the false alarm rate rapidly approaches zero as the threshold increases. This indicates that the proposed S^2CTPL method possesses superior background suppression capability. Overall, the superior ROC performance of the proposed method can be attributed to the effective integration of the spectral-spatial residual extraction module and the multi-directional spatial context-aware module, which enable the extraction of discriminative spatial features. Meanwhile, the prior-guided prototype learning mechanism effectively alleviates spectral variability and enhances the semantic consistency of target representation.

Finally, a quantitative comparison based on multiple AUC metrics is conducted, as summarized in Tables II-VI. Each table reports several evaluation indicators, including AUC_{PD-FAR}, $AUC_{PD-\tau}$, $AUC_{FAR-\tau}$, AUC_{TD}, AUC_{BS}, AUC_{TD-BS}, AUC_{ODP}, AUC_{OD}, and AUC_{SNPR}, where the best and second-best results are highlighted respectively by bolding in red and underlining. For the Urban and Airport data, the proposed S^2CTPL method achieves the best performance in almost all AUC metrics, while obtaining the second-best performance in $AUC_{PD-\tau}$, demonstrating its strong overall detection capability. Similarly, in the RIT Campus dataset, the proposed method achieves optimal results in most metrics, with only a few indicators showing slightly lower performance, indicating its robustness under complex background conditions. In Sandiego, HTD-IRN achieves superior performance in several background-related metrics, such as $AUC_{FAR-\tau}$, AUC_{BS}, and AUC_{SNPR}. This can be attributed to its interpretable module introduced in the encoder-decoder framework, which effectively suppresses background regions with spectral characteristics similar to airplane targets. However, for the remaining detection-related metrics, including AUC_{PD-FAR}, $AUC_{PD-\tau}$, AUC_{TD}, AUC_{TD-BS}, AUC_{ODP}, and AUC_{OD}, the proposed S^2CTPL method achieves the best performance. For the Grand Isle dataset, the proposed method and HTDmamba achieve competitive performance, where the S^2CTPL method obtains optimal results in most key metrics, including $AUC_{PD-\tau}$, AUC_{TD}, AUC_{TD-BS}, AUC_{ODP}, and AUC_{OD}, while maintaining competitive performance in the remaining indicators.

Overall, the consistent superiority of the proposed S^2CTPL method across multiple datasets and evaluation metrics demonstrates its effectiveness for hyperspectral target detection. This advantage stems from the proposed spectral-spatial context-aware feature extraction modules, which capture discriminative spectral-spatial information, as well as the prior-guided prototype learning mechanism, which effectively

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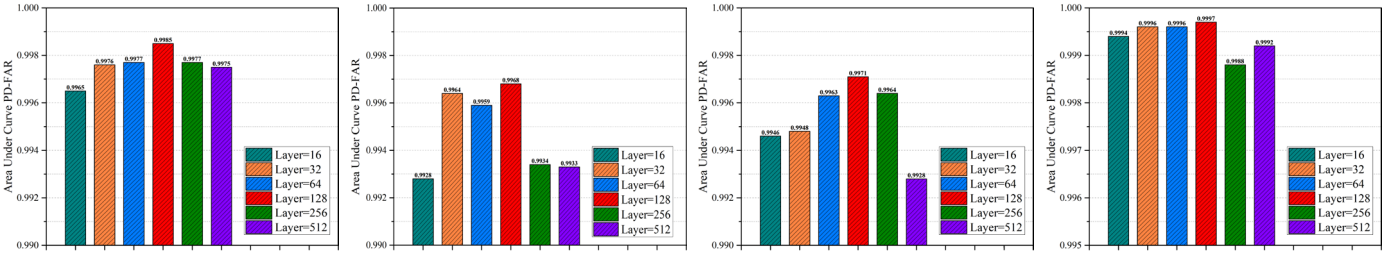


Fig.4. The AUC performance comparison with different numbers of prototypes: From left to right: Urban, Sandiego, Airport, and RIT Campus.

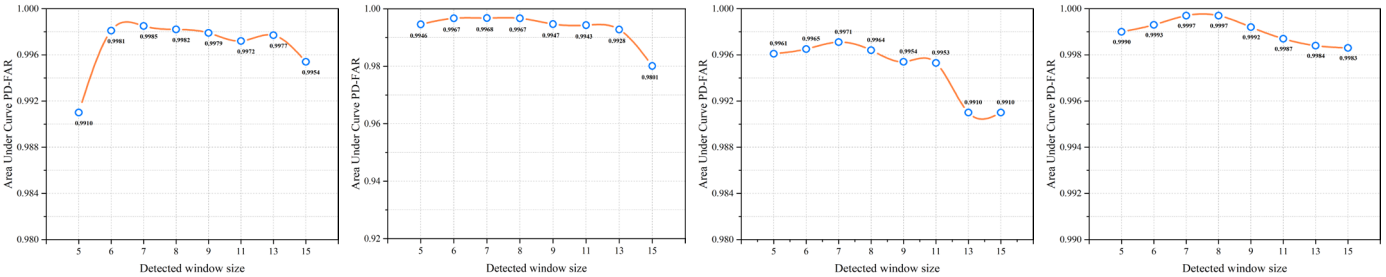


Fig.5. The AUC performance comparison with different window sizes: From left to right: Urban, Sandiego, Airport, and RIT Campus.

TABLE VII
ABLATION STUDY OF DIFFERENT PROTOTYPES ON FOUR HSIS

Module	Urban	Sandiego	Airport	Grand Isle
N prototype	0.9969	0.9395	0.9751	0.9971
R prototype	0.9972	0.9908	0.9931	0.9972
P prototype	**0.9985**	**0.9968**	**0.9971**	**0.9978**

TABLE VIII
ABLATION STUDY OF DIFFERENT LOSSES ON FOUR HSIS

Loss	Urban	Sandiego	Airport	Grand Isle
$\mathcal{L}_{cls}$	0.9969	0.9940	0.9852	0.9922
$\mathcal{L}_{cls}+\mathcal{L}_{que}$	09974	0.9962	0.9877	0.9943
$\mathcal{L}_{cls}+\mathcal{L}_{que}+\mathcal{L}_{div}$	**0.9985**	**0.9968**	**0.9971**	**0.9978**

mitigates spectral variability and enhances target semantic consistency. As a result, the proposed method is capable of suppressing complex backgrounds while accurately detecting target pixels, achieving superior detection performance in challenging hyperspectral scenarios.

D. Parameter Analysis

(1) Analysis of the number of prototypes: In the proposed target prototype learning network, the prototype memory bank is initialized using prior target spectra and is further optimized during training. As a learnable memory structure, the number of prototypes directly determines the diversity of target semantic representations that can be captured, which plays a crucial role in the performance of the proposed method. To investigate the sensitivity to this parameter, the number of prototypes is set to [16, 32, 64, 128, 256, 512], and the corresponding AUC results on the four datasets are illustrated in Fig. 4.

It can be observed that the AUC values initially increase with the number of prototypes and then gradually decrease after reaching a certain point. This phenomenon can be explained as follows. When the number of prototypes is small, the representation capacity of the memory bank is limited, leading to insufficient modeling of target spectral variability and inaccurate prototype estimation. As the number of prototypes increases, the model becomes capable of capturing more diverse target semantics, resulting in improved detection performance. However, when the number of prototypes becomes excessively large, the risk of overfitting increases, especially in HTD tasks where target pixels are limited. Meanwhile, the computational cost also increases significantly.

In summary, the experimental results indicate that setting the number of prototypes to 128 achieves the best and most stable performance across all datasets.

(2) Analysis of the patch size for sample construction: In the proposed sample construction strategy, image patches are used to simulate target occurrences under different background environments. The patch size directly affects both spectral and spatial feature extraction, thereby influencing the learning of target prototypes. To evaluate the impact of patch size, experiments are conducted with patch sizes of [5×5, 6×6, 7×7, 8×, 9×9, 11×11, 13×13, 15×15]. The corresponding AUC results are presented in Fig. 5.

It can be observed that when the patch size is small, the AUC values are relatively low. As the patch size increases, the AUC values gradually improve, but begin to decrease when the patch size becomes excessively large. This trend can be explained by two aspects. First, when the patch size is small, the background region is limited, making it insufficient to simulate diverse background conditions under which targets may appear. In addition, the extraction of spectral-spatial features is constrained, leading to suboptimal representation learning. On the other hand, when the patch size is too large, the background becomes more complex and introduces additional interference, which negatively affects both feature extraction and prototype learning, ultimately degrading detection performance.

Overall, the S^{2}CTPL method shows relatively low sensitivity to this parameter when the patch size is smaller than 10×10. However, performance degrades rapidly when the patch size becomes too large. Based on the experimental results, a patch size of 7×7 achieves the most robust performance.

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This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3749396

E. Ablation Study

(1) Ablation study on prototype memory bank: In the target prototype learning network, the prototype memory bank serves as a learnable structure to capture diverse target semantic representations, which is essential for the overall performance of the S^{2}CTPL. To investigate the impact of different initialization strategies, three configurations are considered: without a prototype memory bank (N prototype), with a randomly initialized prototype memory bank (R prototype), and with a prototype initialized using target prior spectra (P prototype). The AUC results conducted on four HSIs, i.e., the Urban, the Sandiego, the Airport, and the Grand Isle, are reported in Table VII.

From the results, it can be observed that the absence of a prototype memory bank leads to the lowest detection performance among all datasets, particularly in the Sandiego and the Airport dataset. This indicates that the model lacks sufficient capability to represent diverse target characteristics without the support of prototype learning. When a randomly initialized prototype memory bank is introduced, the AUC values increase among all datasets, demonstrating that the prototype memory mechanism itself contributes positively to HTD performance. However, the performance is still inferior compared to the prior-guided initialization strategy. When the prototype memory bank is initialized using prior spectra, the S^{2}CTPL detector achieves the best performance on all data. This improvement can be attributed to the fact that prior target spectra provide a strong initialization for prototype estimation, enabling the network to more accurately learn representative target prototypes at an early stage of training. As a result, the semantic consistency of target representation is better preserved during the detection process, leading to enhanced detection accuracy.

(2) Ablation study on Loss Functions: The S^{2}CTPL method incorporates three loss functions, including the binary cross-entropy (BCE) loss for basic target-background classification ($\mathcal{L}_{cls}$), the discriminative query loss for enhancing feature discrimination ($\mathcal{L}_{que}$), and the prototype diversity loss for encouraging diversity among target prototypes ($\mathcal{L}_{div}$). Among them, the $\mathcal{L}_{cls}$ loss serves as the fundamental loss function and is indispensable for the proposed method. To estimate the contribution of each loss function, experiments are conducted using different combinations of these loss functions. The AUC results conducted on four HSIs, i.e., the Urban, the Sandiego, the Airport, and the Grand Isle, are summarized in Table VIII.

From the results, it can be observed that when only the $\mathcal{L}_{cls}$ loss is used, the detection performance is the lowest across all datasets. After incorporating the $\mathcal{L}_{que}$ loss and the $\mathcal{L}_{div}$ loss, the AUC values are consistently improved, demonstrating the effectiveness of the proposed loss design. Specifically, the $\mathcal{L}_{que}$ loss enhances the ability to distinguish between target and background features, thereby improving detection accuracy. Meanwhile, the $\mathcal{L}_{div}$ loss encourages the learning of diverse target prototypes within the memory bank, which strengthens the semantic consistency of target representation. The combination of these loss functions improves the detection capability of the prior-guided prototype learning framework.

IV. Conclusion

In this paper, we proposed a novel spectral-spatial context-aware target prototype learning framework, termed S^{2}CTPL, for HTD. To address the challenges caused by limited target samples and significant spectral variability, a context-aware sample construction strategy was first developed to generate diverse and physically plausible samples. Subsequently, a hierarchical spectral-spatial feature extraction network was designed, where the SSREM module effectively captures discriminative local features, and the MSCAM models long-range spatial dependencies within hyperspectral patches. Furthermore, a prior-guided prototype learning mechanism was introduced to represent targets with multiple learnable prototypes, enabling the model to better handle intra-class spectral variations of targets and enhance target-background separability. Extensive experiments conducted on multiple benchmark HSI datasets demonstrate that the S^{2}CTPL method achieves superior performance compared with several state-of-the-art detectors in terms of both detection accuracy and robustness. In particular, the proposed method shows strong capability in suppressing complex backgrounds and maintaining stable detection performance.

In future work, we will explore more efficient prototype learning strategies and lightweight network architectures to further improve computational efficiency and extend the proposed framework to more challenging real-world scenarios.

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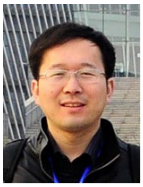
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This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3749396



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